HORIZON

SOVEREIGN AUTONOMOUS ENTERPRISE INFRASTRUCTURE RECOVERY PLATFORM

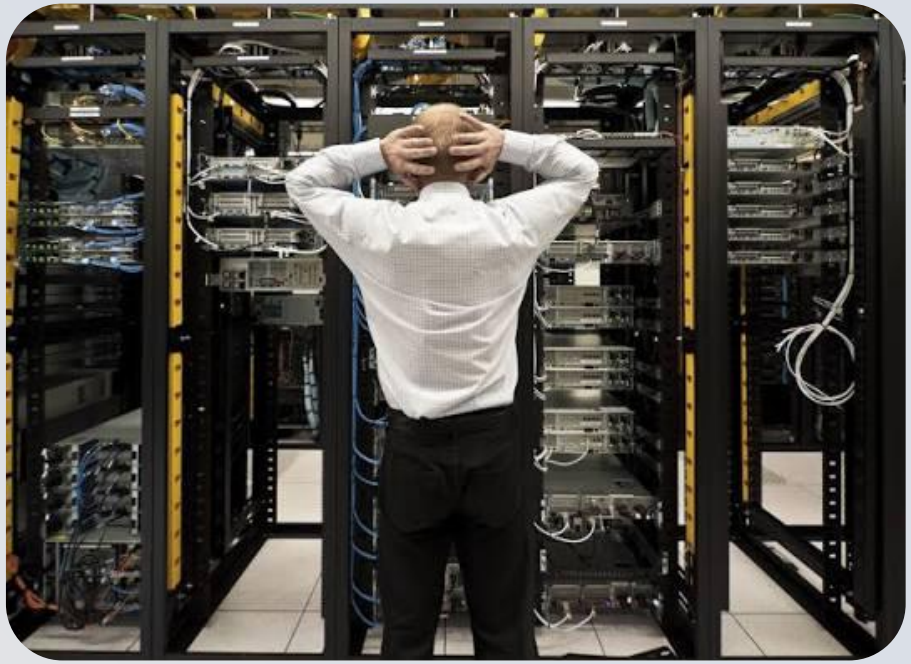
THINK PAGERDUTY PLUS AN AUTOPILOT, BUILT IN INDIA, WITH PROOF OF EVERY ACTION

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Overview

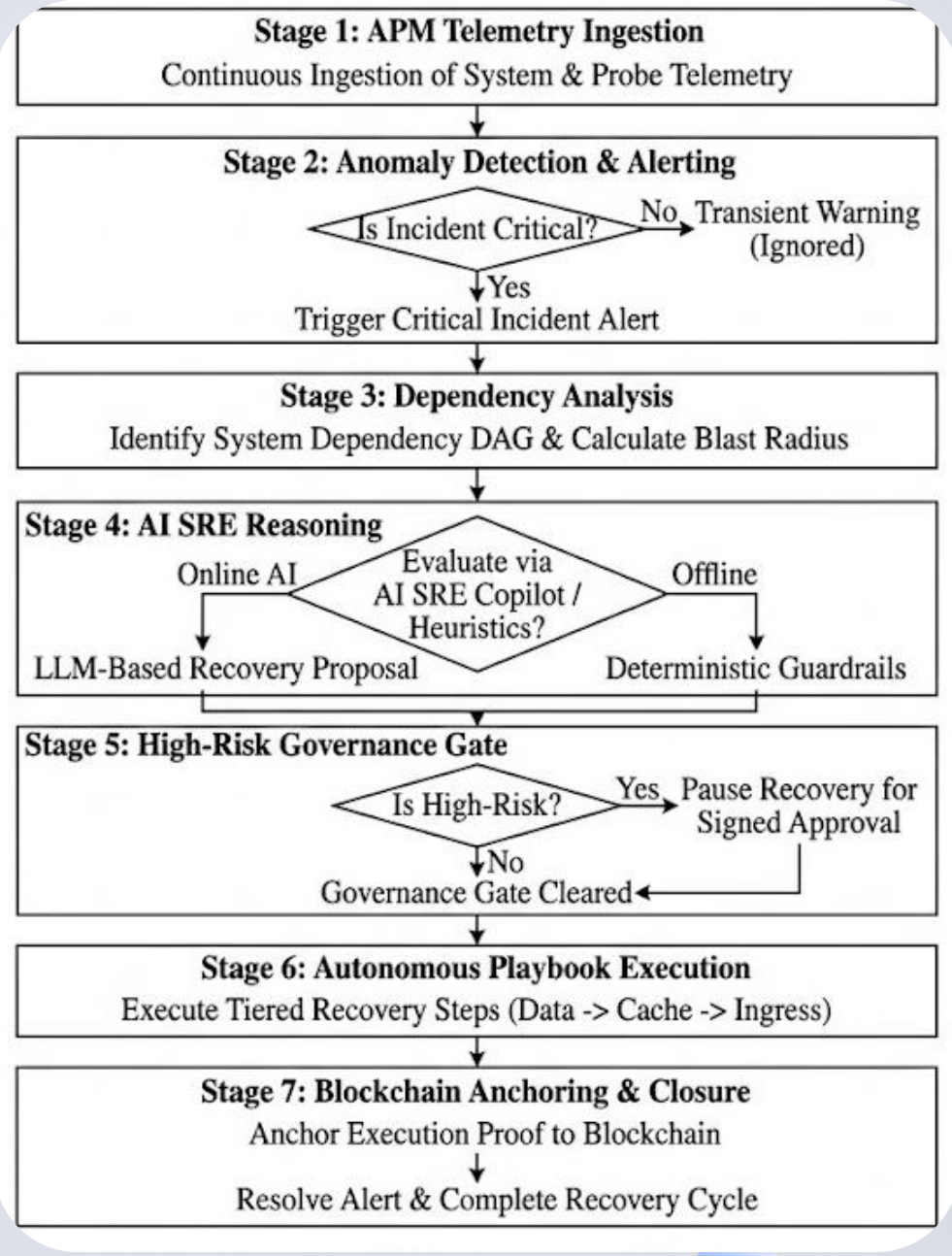
- **Problem:** Distributed systems fail in chains, while manual recovery and incorrect restart sequences extend costly downtime.
- **Idea:** Detect failures, map dependencies, and autonomously restore services in the mathematically correct order.
- **USP:** Combines graph-based recovery, human approval for high-risk actions, and blockchain-backed proof of every action.
- **Innovation:** Deterministic recovery algorithms, multilingual AI assistance, and tamper-evident audit trails in one platform.

HORIZON



Technical Approach

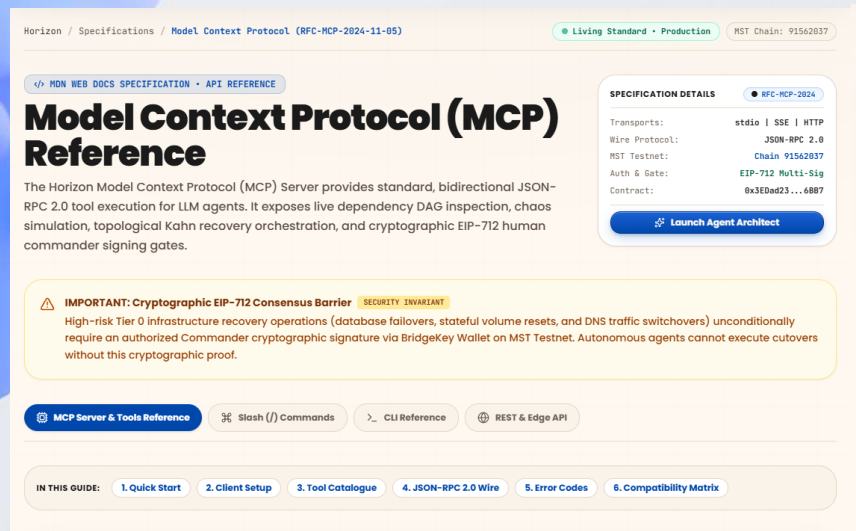
- Maintains live service DAGs in ClusterStateManager and confirms outages using a 3-consecutive-miss probe to eliminate flapping.
- Traverses reverse dependency edges via computeBlastRadius in O(V + E) time to isolate the exact downstream cascade before acting.
- Derives bottom-up recovery order via getTopologicalOrder (Databases → Caches → APIs) to eliminate thundering-herd crash loops.
- Runs low-risk heals automatically while pausing hazardous mutations for signed wallet approvals via mstBlockchain.
- Verifies node health in Pipeline Deployer before unlocking dependants and anchors SHA-256 Merkle proofs to HorizonAuditVault.

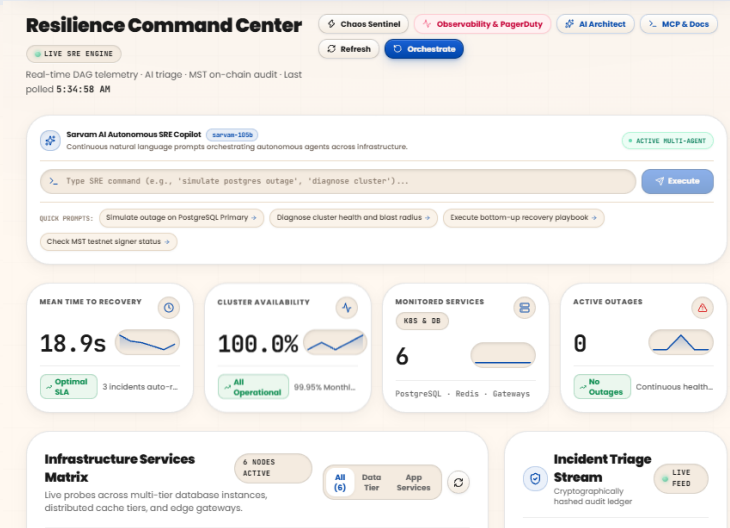


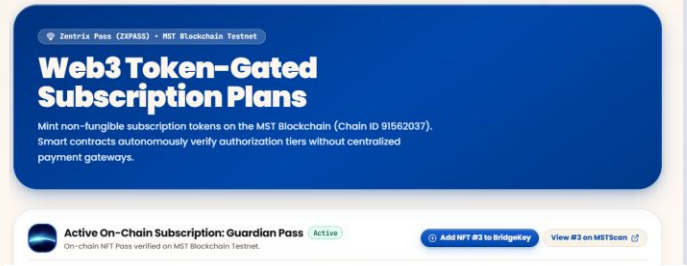
Working Deployed Prototype

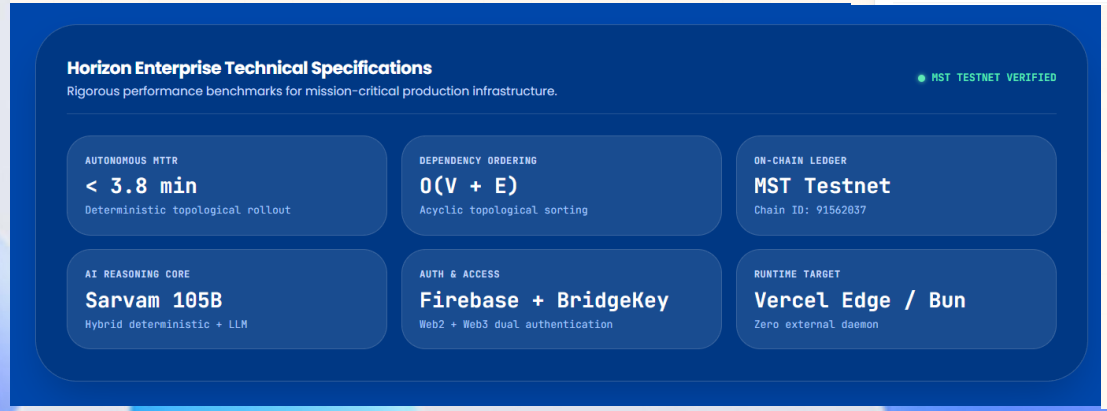
Deployed Link: horizon-aiops.vercel.app

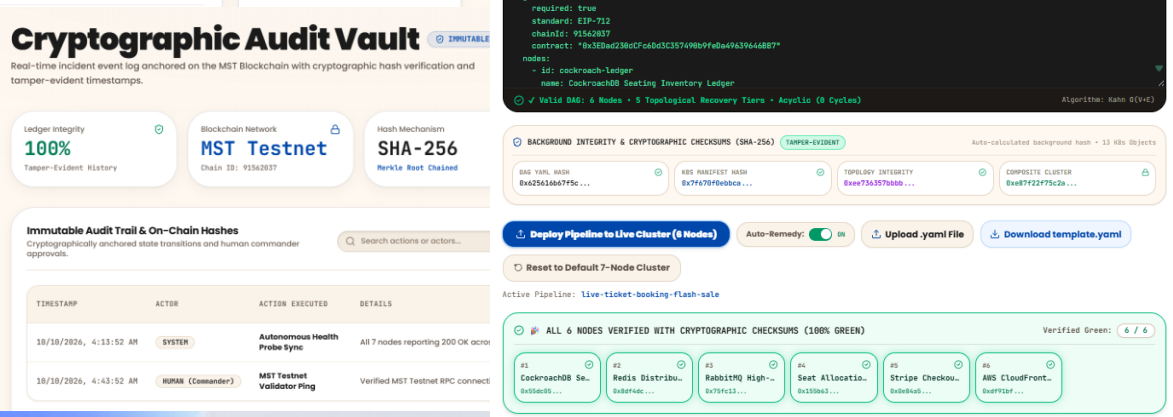
Github Repo: [Horizon Repository]











Feasibility and Viability

- O(V + E) Scalability: Computes 1,000+ node recovery in < 3 ms via DependencyGraph without heavy compute.
- Deterministic Core: Executes recovery via pure TypeScript in PipelineDeployer, isolated from LLM hallucinations.
- Native Connectors: Plugs directly into Docker via dockerBridge, Kubernetes, and standard MCP IDE tools. Cycle & Flap Resilient: Built-in DFS cycle detection stops deadlocks, while sliding-window probes prevent flapping.
- Low-Cost Web3: Hardware-backed EIP-712 wallet gates in mstBlockchain anchor on-chain for < ₹0.10 gas.

- High ROI: Slashes $50k/min downtime losses by cutting MTTR from 45 mins to < 2 minutes.
- Regulatory Driver: Automates India's mandatory 6-hour CERT-In reporting via on-chain HorizonAuditVault.
- 80%+ Gross Margins: High-margin SaaS model with < ₹12 marginal compute and gas cost per recovery.
- PLG Flywheel: Developers adopt @horizon/mcp-server inside Cursor and Claude for organic bottom-up sales.
- Sovereign Advantage: INR pricing removes USD forex volatility, with 100% Indian data residency for BFSI.

Marketing strategies

- Datadog / Dynatrace: Only alerts on issues; Horizon actuates automated recovery.
- PagerDuty / Rootly: Pages on-call humans; Horizon resolves incidents autonomously.
- StackStorm / Rundeck: Runs static scripts; Horizon derives dynamic DAG plans.
- Generic LLM Agents: Hallucinates commands; Horizon enforces deterministic math.
- Patent US12556441B1: Static mapping; Horizon re-plans on health probe failures.

Business Model Canvas

- Core Value: Slashes MTTR by 95%, eliminates out-of-order crash loops, and guarantees tamper-proof audit trails.
- Target Market: Indian cloud scale-ups, regulated FinTech/Neobanks (CERT-In 6h compliance), and HealthTech.
- Revenue Model: Predictable INR SaaS subscriptions (₹0 to ₹24,999/mo) + high-ticket Enterprise annual contracts.
- Unit Economics: 82% gross margins with < ₹12 marginal serverless and blockchain cost per recovered incident.

Thank you

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